

Components required to create straight line

- x and y point
- Slope
- Intercept

Components required to create straight line

- $x = 1$
- $y = 1$
- Slope = 1.5

$$y = mx + c$$

Slope

$$m = \frac{\text{Change in Y}}{\text{Change in X}}$$

$$\text{Slope } (m) = \frac{Y_2 - Y_1}{X_2 - X_1}$$

X Y

2 10

4 20

$$Y_1 = 10$$

$$Y_2 = 20$$

$$X_1 = 2$$

$$X_2 = 4$$

$$m = \frac{20 - 10}{4 - 2}$$

$$m = \frac{10}{2}$$

$$m = 5$$

Equation of Straight Line

$$y = m x + c$$

m (Slope or Coefficient) = slope of the line
c Intercept (where the best-fit line meets the Y axis).

$$M=1.3$$

$$C=0$$

X	m x + c = y
1.0	(1.3 * 1) + 0 = 1.3
3.0	(1.3 * 3) + 0 = 3.9
4.0	(1.3 * 4) + 0 = 5.2
6.0	(1.3 * 6) + 0 = 7.8
7.0	(1.3 * 7) + 0 = 9.1
8.0	(1.3 * 8) + 0 = 10.4

$$M=1.2$$

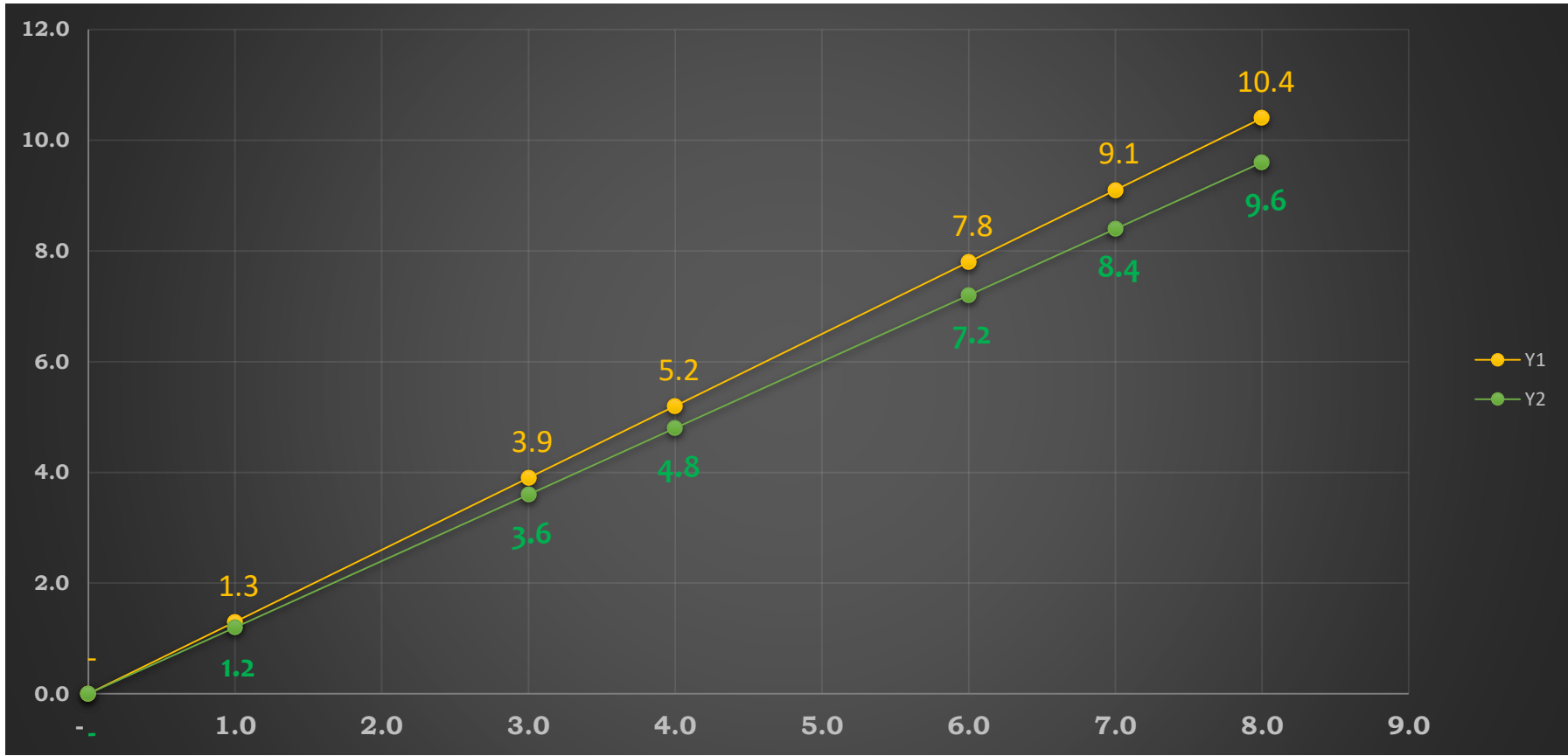
$$C=0$$

m x + c = y
(1.2 * 1) + 0 = 1.2
(1.2 * 3) + 0 = 3.6
(1.2 * 4) + 0 = 4.8
(1.2 * 6) + 0 = 7.2
(1.2 * 7) + 0 = 8.4
(1.2 * 8) + 0 = 9.6

M=1.3
C=0

x	y_1	Y_2
1.0	1.3	01.2
3.0	3.9	3.6
4.0	5.2	4.8
6.0	7.8	7.2
7.0	9.1	8.4
8.0	10.4	9.6

M=1.2
C=0



Equation of Straight Line

$$y = \mathbf{m} \mathbf{x} + \mathbf{c}$$

$$y = \mathbf{m} \mathbf{x} + \mathbf{c}$$

Equation of Straight Line

$$ax + by + c = 0$$

$$by + c = -ax$$

$$by = -ax + (-c)$$

$$y = \frac{-\mathbf{a}}{\mathbf{b}} \mathbf{x} + \frac{-\mathbf{c}}{\mathbf{b}}$$

$\mathbf{m}$ OR $\frac{-\mathbf{a}}{\mathbf{b}}$ = (Slope or Coefficient) slope of the line

$\mathbf{c}$ OR $\frac{-\mathbf{c}}{\mathbf{b}}$ = Intercept (where the best-fit line meets the Y axis).

$$-1x + 2y - 2 = 0$$

$$-1x + 2y - 2 = 1$$

$$-1x + 2y - 2 = -1$$

$$-1x + 2y - 3 = 0$$

$$-1x + 2y - 1 = 0$$

$$y = m x + c$$

$$-1x + 2y - 2 = 1$$

$$-1x + 2y - 3 = 0$$

$$-1x + 2y - 2 = 1$$

$$-1x + 2y - 3 = 0$$

$$-1x + 2y = 1 + 2$$

$$-1x + 2y = +3$$

$$-1x + 2y = 3$$

$$2y = 1x + 3$$

$$2y = 1x + 3$$

$$y = \frac{1}{2} x + \frac{3}{2}$$

$$y = \frac{1}{2} x + \frac{3}{2}$$

$$y = 0.5 x + 1.5$$

$$y = 0.5 x + 1.5$$

$$-1x + 2y - 2 = -1$$

$$-1x + 2y = -1 + 2$$

$$-1x + 2y = 1$$

$$2y = 1x + 1$$

$$y = \frac{1}{2}x + \frac{1}{2}$$

$$y = 0.5x + 0.5$$

$$-1x + 2y - 1 = 0$$

$$-1x + 2y = 1$$

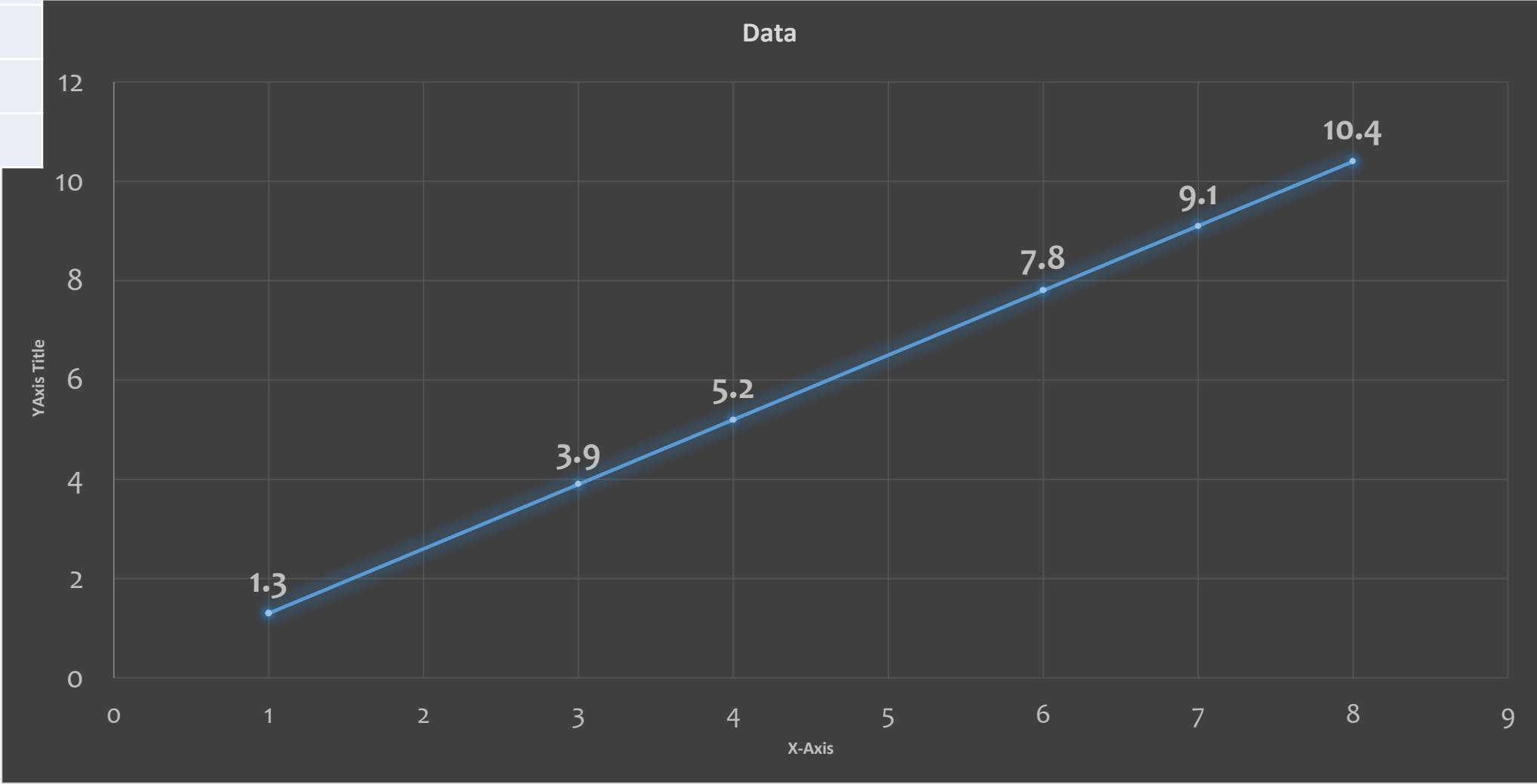
$$2y = 1x + 1$$

$$2y = 1x + 1$$

$$y = \frac{1}{2}x + \frac{1}{2}$$

$$y = 0.5x + 0.5$$

X	Y
X = 1	y = 1.3
X = 3	y = 3.9
X = 4	y = 5.2
X = 6	y = 7.8
X = 7	y = 9.1
X = 8	y = 10.4



Equation of Straight Line

$$ax + by + c = 0$$

$$y = m x + c$$

$$X=[1,3,4,6,7,8]$$

If

$$a = 1.3$$

$$b = -1$$

$$c = 0$$

Equation of Straight Line

$$ax + by + c = 0$$

$$1.3(1) + (-1)y + 0 = 0$$

$$1.3 - y + 0 = 0$$

$$1.3 - y = 0$$

$$-y = 0 - 1.3$$

$$-y = -1.3$$

$$y = 1.3$$

x	(ax+by+c=0) (Substitute values)	Output
1	$1.3(1) - y + 0 = 0$	$(y = 1.3)$
3	$1.3(3) - y + 0 = 0$	$(y = 3.9)$
4	$1.3(4) - y + 0 = 0$	$(y = 5.2)$
6	$1.3(6) - y + 0 = 0$	$(y = 7.8)$
7	$1.3(7) - y + 0 = 0$	$(y = 9.1)$
8	$1.3(8) - y + 0 = 0$	$(y = 10.4)$